

École Nationale des Sciences Appliquées de Berrechid

Génie Mécatronique et Systèmes Intelligents

## End-of-Year Internship Project

Final internship report

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# Integrated Brake-by-Wire and Steer-by-Wire Control Allocation

A fault-tolerant vehicle simulation demonstrator

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ENSA Berrechid – SEGULA Technologies internship framework

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# Dedication

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To our families, for their encouragement throughout our studies.

To our teachers, whose work has helped us build the scientific and technical foundations of this project.

To our friends and fellow students, for the discussions, mutual support and shared desire to learn.

Laaraki Mohamed Amine  
Aymen Bentaleb

# Acknowledgements

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We thank SEGULA Technologies and ENSA Berrechid for the academic and professional framework in which this internship project was developed. Working on integrated braking and steering control offered an opportunity to connect mechatronics, vehicle dynamics, numerical optimization and engineering verification.

We express our gratitude to Mr Elmehdi Chokri, project guarantor, for his guidance and support throughout our work on this simulation demonstrator.

We warmly thank our academic supervisors, Pr Said Ben Alla and Pr Issam Amellal, and the teaching staff of ENSA Berrechid for their role in our training and the academic supervision of this work.

Our thanks also go to those who will evaluate this report, and to everyone who supported us during the project. This report is an opportunity to share both the engineering solution developed and the lessons learned from its implementation.

# Abstract

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In this end-of-year internship project, we develop a functional simulation demonstrator for integrated Brake-by-Wire and Steer-by-Wire control allocation. Four wheel brakes and a shared steering rack driven by two independent functional channels are coordinated by a constrained quadratic program. The controller redistributes longitudinal force and yaw-moment requests according to diagnosed actuator availability, while a nonlinear 13-state vehicle model represents tire saturation, combined force capacity, load transfer and actuator dynamics. Physically unattainable demands are treated through bounded best-effort allocation.

The completed demonstrator is supported by 45 MATLAB unit tests, five nominal maneuvers and 30 original fault scenarios. Differential-Braking Backup Steering (DBBS) satisfies the project pulse criteria in all eight assessed cases: worst path deviation is 0.295 m, worst yaw-rate RMSE is 0.0115 rad/s and maximum deceleration is 0.277 g. In the evaluated front-brake-loss authority maneuvers, at least 61.1% of the nominal upper-bound braking capability is retained, with peak sideslip limited to 0.0605 rad. Targeted 1 ms refinement confirms the principal results.

The final academic scope is a reproducible, model-based demonstration at a documented reference operating condition, complemented by a qualitative safety analysis. Broader operating-envelope guarantees, abrupt faults during fully saturated braking and hardware validation are organized as future perspectives. The numerical records are preserved, and completion of the academic demonstrator is distinguished from production certification or an all-condition safety claim.

**Keywords:** BbW, SbW, DBBS, control allocation, quadratic programming, vehicle dynamics, fault tolerance, MATLAB.

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# Abbreviations and notation

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|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| BbW / SbW       | Brake-by-Wire / Steer-by-Wire                               |
| DBBS            | Differential-Braking Backup Steering                        |
| QP              | Quadratic program                                           |
| RMSE            | Root mean square error                                      |
| FMEA / AMDEC    | Failure modes and effects analysis                          |
| FTA / DFA       | Fault-tree / dependent-failure analysis                     |
| ASIL / ECU      | Automotive Safety Integrity Level / Electronic Control Unit |
| CDC / PFA       | Project specification / end-of-year project                 |
| $F_x, F_y, F_z$ | Longitudinal, lateral and normal forces (N)                 |
| $M_z, r, \beta$ | Yaw moment (N m), yaw rate (rad/s), sideslip (rad)          |
| $\delta, \mu$   | Road-wheel/rack angle (rad), friction coefficient           |
| $h_p, h_k$      | Physical and diagnosed health masks                         |

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# General introduction

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Electrically commanded braking and steering create a coordination problem as well as an actuator-control problem. When one effector becomes unavailable, a command that was feasible with the complete actuator set may no longer be attainable. A useful degraded response therefore requires explicit prioritization, physical limits and a method for exploiting the effectors that remain available.

We address this problem by treating the brake and steering subsystems as one over-actuated plant. Steering and differential braking can both influence yaw, but they are not interchangeable: braking reduces speed and shares the tire friction budget with lateral force. This coupling motivates a nonlinear vehicle model and a constrained allocation strategy.

We follow a progressive engineering sequence: formalize the functional architecture, verify the plant without allocation, connect nominal control, introduce diagnosed channel losses, improve degraded behavior, then preserve and independently audit the results. The deliverable is a working simulation system accompanied by a reproducible evidence set and this bilingual report.

In the following chapters, we present the host organization and project objectives, the literature review, the functional architecture, the mathematical model, the allocator, our simulation results, the safety rationale, and future perspectives. We evaluate our demonstrator within the operating conditions and scenario set defined in Chapter 1; broader industrial validation remains a subsequent development stage.

# Host organization and project framing

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## 1.1 SEGULA Technologies and the Moroccan context



Figure 1.1 – SEGULA Technologies, host engineering context.

SEGULA Technologies is an engineering group. Its automotive activities include development, testing and technology consulting for vehicle manufacturers and suppliers. The Moroccan organization presents automotive engineering through its Casablanca design office and aerospace activities through SIMRA. Its official website also lists locations in Agadir and Tanger. These elements situate the internship within an engineering and industrial-development environment [12, 13].

Our project concerns control engineering and numerical demonstration. It does not claim integration into a named customer's production vehicle, access to proprietary vehicle data or validation by an industrial vehicle program. The reference vehicle is assembled from published and explicitly assumed parameters.

## 1.2 Academic organization

Table 1.1 – Project identification.

|                       |                                              |
|-----------------------|----------------------------------------------|
| Institution           | ENSA Berrechid                               |
| Programme             | Génie Mécatronique et Systèmes Intelligents  |
| Prepared by           | Laaraki Mohamed Amine; Aymen Bentaleb        |
| Academic supervisors  | Pr Said Ben Alla; Pr Issam Amellal           |
| Project guarantor     | Elmehdi Chokri                               |
| Reference environment | MATLAB R2026a Update 5; Optimization Toolbox |
| Deliverable           | Simulation demonstrator and FR/EN report     |

## 1.3 Project objectives and scope

Our objective is to develop and document a functional simulation demonstrator for integrated braking and steering control. We use the CDC to guide our architecture and quantitative

benchmarks, and assess the system on a reference vehicle model and a defined set of nominal and degraded maneuvers. Our work covers modeling, constrained allocation, selected actuator losses and analysis of residual performance. The table below distinguishes the capabilities we demonstrate from the developments we identify as future perspectives.

Table 1.2 – Completed deliverable and future development boundary.

| Current deliverable                                      | Future perspectives                                                                   |
|----------------------------------------------------------|---------------------------------------------------------------------------------------|
| Nonlinear reference vehicle; common rack and four brakes | Vehicle-specific identification and experimental correlation                          |
| Nominal allocation; selected fail-silent channel losses  | Stuck actuators, sensor/ECU/common-cause faults                                       |
| Measured braking and DBBS performance in the catalog     | Abrupt faults during fully saturated braking; continuous recovery/envelope guarantees |
| Traceable simulation and qualitative safety rationale    | Hardware independence, real-time timing, certification                                |

Appendix C gives the clause-by-clause status of the engineering baseline, so that demonstrated results are never read as blanket compliance. We preserve the simulation records so that each result remains traceable to its test conditions and controller version. Our additional investigations of saturated transitions inform the future-work chapter. Throughout this report, we explain the demonstrated behavior, our design choices and the practical next steps for the system.

## 1.4 Method and project progression

Table 1.3 – Engineering work sequence.

| Stage | Question                                    | Produced evidence                                  |
|-------|---------------------------------------------|----------------------------------------------------|
| 1     | What is redundant?                          | Architecture and fault semantics                   |
| 2     | Does the plant behave consistently?         | Open-loop and actuator benches                     |
| 3     | Can demands be allocated nominally?         | Five time-varying maneuvers                        |
| 4     | What remains after a channel loss?          | 30 scenarios and residual authority                |
| 5     | How can degraded behavior improve?          | Rear reserve and DBBS feedback tuning              |
| 6     | Can the result be explained and reproduced? | Independent audits, source snapshots, final report |

We carried out most of this work together, from the literature review and design to result analysis and report writing. Our contributions are joint.

# Literature review and design choices

---

We examine how redundant vehicle actuators can preserve braking and lateral control after a fault. Our review connects three questions: which physical mechanisms provide backup authority, how that authority can be allocated under constraints, and which assumptions are necessary before claiming fault tolerance. The references identified during Gate 1 inform our design; their vehicle architectures and test conditions are distinguished from our own.

## 2.1 Hardware redundancy and cross-domain compensation

Duplicating an actuator preserves a familiar control mechanism, provided that the surviving branch has sufficient capability and is not defeated by the same cause. Cross-domain compensation instead uses another vehicle function to create a useful force or moment. The two approaches can coexist: our steering drives provide local redundancy, while differential braking provides a second means of influencing yaw.

Eckstein’s account of the SpeedE research vehicle describes wheel-individual steering and cross-domain compensation. A faulty steering intervention can be counteracted by the other steered wheel together with rear-axle torque vectoring [2]. The important precedent is reasoning at vehicle-function level rather than equating redundancy with identical duplicated components.

We retain this functional principle, but our plant has one common rack and no independently commanded propulsion torques. It cannot counter-steer one front wheel against the other. Its available redundancy must therefore be assessed using its own four brake-force channels and shared steering angle. Results obtained with wheel-individual steering cannot be transferred directly to this architecture.

## 2.2 Braking as a backup for steering

Two different mechanisms appear under the broad expression “steering by braking”. Dominguez-Garcia, Kassakian and Schindall propose brake-actuated steering in which asymmetric wheel forces actuate the steering mechanism after mechanical decoupling from the failed primary actuator [3]. This approach depends on steering geometry and the mechanical freedom of the rack. It requires a model of steering torque and motion.

Kissai, Mouton and Monsuez study EPAS backup through differential braking: separate wheel braking produces a vehicle yaw moment, with allocation coordinating the available brakes [4]. Their weighted least-squares formulation provides a useful precedent for bounded allocation. Its relevance to our work is the explicit management of brake authority, rather than an assumption that braking recreates every capability of the steering system.

We adopt differential-braking backup steering (DBBS) in this second sense. When both steer-

ing drives are lost, the model holds the existing rack angle and the allocator creates additional yaw through brake asymmetry. We do not model brake-induced rack motion. This distinction fixes what our simulations can establish: a controlled lateral response with deceleration, under a specified rack state and tire-force budget.

### 2.3 Steering assistance after brake loss

Jiang and colleagues address single-wheel electromechanical-brake failure through coordinated brake-torque redistribution and active front steering [5]. Their treatment of the conflict between deceleration and lateral stability is directly relevant to our front-brake-loss cases. Brake utilization and lateral support cannot be selected independently when both consume the same tire adhesion.

Our response is to retain a larger rear lateral-force reserve after diagnosed front-brake loss. This is a project-specific control choice evaluated with our nonlinear plant, not a reproduction of the published controller or its numerical results. The comparison explains why a small reduction in braking utilization can be justified when it substantially reduces sideslip.

Han and colleagues investigate steering-system failure in four-wheel independent by-wire steering electric vehicles [6]. Gate 1 identified their hierarchical predictive-control and torque-distribution approach as another example of actuator coordination. We classify it here as related steering-fault work, not evidence that steering alone restores lost braking. Its additional steering and drive degrees of freedom differ from our common-rack, braking-only allocation.

### 2.4 Control allocation methods

The survey by Johansen and Fossen separates the motion-control demand, its allocation among effectors, and local actuator control [1]. This hierarchy is useful for our implementation: generalized force and yaw moment remain the interface between vehicle-level objectives and actuator-level constraints. The following comparison explains our selection; it is not an experimental ranking of algorithms.

Table 2.1 – Allocation alternatives and implications for our design.

| Method           | Useful property                                   | Design consideration                                                                   |
|------------------|---------------------------------------------------|----------------------------------------------------------------------------------------|
| Pseudoinverse    | Direct solution of a linear force mapping         | Clipping alone can leave avoidable tracking error; redistribution requires added logic |
| Daisy chain      | Explicit priority between actuator groups         | The group order determines how remaining authority is used                             |
| Bounded WLS / QP | Weighted error and actuator bounds in one problem | Requires scaling, numerical checks and defined fallback behavior                       |

We select a quadratic program with a locally affine force mapping, bounds and command-rate limits. Force and yaw errors are scaled before weighting so that units do not set the priority implicitly. When a demand exceeds available authority, minimizing a bounded residual

preserves useful actuation. The nonlinear plant remains the reference for actual delivered force and moment; minimizing the affine allocation error is not itself a nonlinear stability proof.

## 2.5 Positioning of our contribution

Our contribution is the integration and verification of these principles in one reproducible academic demonstrator. We combine explicit fault masks, independent physical and diagnosed fault timing, a two-drive common-rack model, constrained best-effort allocation and nonlinear evaluation. We then use the observed braking/sideslip trade-off and DBBS path error to improve the degraded controller.

The reviewed work supports the choice of mechanisms and methods, but does not establish our numerical performance. Our own results are presented in Chapter 6. Likewise, ASIL D motivates the safety analysis without turning simulated redundancy into demonstrated hardware independence. The FMEA and dependency review connect our tested fault consequences to the additional work required for a physical system.

# Functional architecture and control objectives

---

## 3.1 Integrated actuation chain

The control chain comprises a maneuver reference, a yaw-rate servo, a QP allocator, actuator dynamics and the nonlinear plant. Plant state feedback updates allocation at each 2 ms sample. The plant uses the physical fault mask; allocation uses the delivered diagnosis mask. This separation exposes the period during which a failed actuator is still believed available.

$$u = [F_{x,FL}, F_{x,FR}, F_{x,RL}, F_{x,RR}, \delta_c]^T, \quad h = [h_{FL}, h_{FR}, h_{RL}, h_{RR}, h_A, h_B]^T.$$

There is one rack angle, not two angles to sum. Two independently masked inner velocity loops contribute to the same rack motion. The healthy pair is limited to 400 degrees/s; each drive is assigned 240 degrees/s and the full angle range. These are functional design assumptions rather than demonstrated torque capacity.

| Fault                 | Implemented physical effect                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------|
| One brake             | Contact force removed immediately. Latent actuator state decays but cannot generate tire force.   |
| One steering drive    | Its rack-velocity contribution is removed. The survivor retains its former share until diagnosis. |
| Both drives           | Rack holds its current angle. Brakes generate backup yaw.                                         |
| Power / communication | Equivalent to a mask only if the real effect is local, isolated and fail-silent.                  |

A held rack does not represent a floating rack, arbitrary jam or opposing motor torque. Lost/corrupt driver requests, sensor failures and common ECU faults also require different models. Both steering drives unavailable can be two independent losses or one shared-path failure: the physical architecture must decide. Six software flags alone do not demonstrate six independent hardware domains.

## 3.2 Control objectives and fault timing

The requested body acceleration and yaw response are translated into generalized force and moment. The allocator does not directly command a vehicle path; it tracks these quantities using the available actuators. DBBS means yaw generation through left/right braking asymmetry. It

does not refer to steering the rack through scrub-radius effects.

$$F_{x,d} = m(a_{x,ref} - v_y r), \quad M_{z,d} = I_z \left( \dot{r}_{ref} + \frac{r_{ref} - r}{\tau_r} \right).$$

Physical loss occurs at  $t_p$ . The diagnosis becomes available at  $t_f = t_p + t_d$ , sampled on the controller grid. Using separate masks prevents the controller from reacting to a future fault. Diagnosis delays of 0, 10 and 50 ms are imposed as sensitivity cases, not measured detector performance.

### 3.3 Benchmarks used in the demonstrator

Table 3.1 – Scenario evaluation benchmarks.

|                             |                                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| Nominal tracking            | $\text{RMSE}(F_x) \leq 250 \text{ N}; \text{RMSE}(r) \leq 0.012 \text{ rad/s}$ |
| DBBS yaw pulse              | $ r_{ref}  \leq 0.15 \text{ rad/s}; \text{RMSE}(r) \leq 0.03 \text{ rad/s}$    |
| DBBS path / deceleration    | $ e_{lat} _{\max} \leq 0.50 \text{ m}; a_{brake} \leq 0.35g$                   |
| Residual braking plateau    | $a_{brake} \geq 0.60\mu g;  M_z  \leq 567.137 \text{ N m}$                     |
| Operating-domain screen     | $v_x > 0.5 \text{ m/s};  \beta  \leq 0.15 \text{ rad}$                         |
| Mask / recovery diagnostics | 10 ms; 10% band, 50 ms entry and 100 ms dwell                                  |

The braking yaw limit is 5% of a feasible nominal yaw witness and is conservative for this benchmark. Recovery uses an affine reference and is reported as a diagnostic. The final academic scope does not promote these sampled tests into an all-condition physical guarantee.

Table 3.2 – Link between engineering questions and report evidence.

| Question                       | Evidence                                              | Interpretation                                                         |
|--------------------------------|-------------------------------------------------------|------------------------------------------------------------------------|
| Are commands bounded?          | QP bounds, rate logs and unit tests                   | Nominal and catalog checks; saturated transitions discussed separately |
| Does allocation react to loss? | Separate physical/known masks and saved timing traces | Imposed diagnosis delay, not detector qualification                    |
| What authority remains?        | Braking plateaus, DBBS pulses and sampled witnesses   | Quantified at stated states and maneuvers                              |
| What does redundancy cover?    | Full FMEA and dependency register                     | Functional rationale; hardware independence requires further evidence  |

### 3.4 Implemented control chain

Figure 3.1 reconstructs the functional structure of the executed MATLAB scripts. Each block corresponds to a named function or a model stage. It is an implementation diagram, not a screenshot of the historical Simulink prototype. Solid arrows carry control or state signals; dashed arrows carry imposed health information.

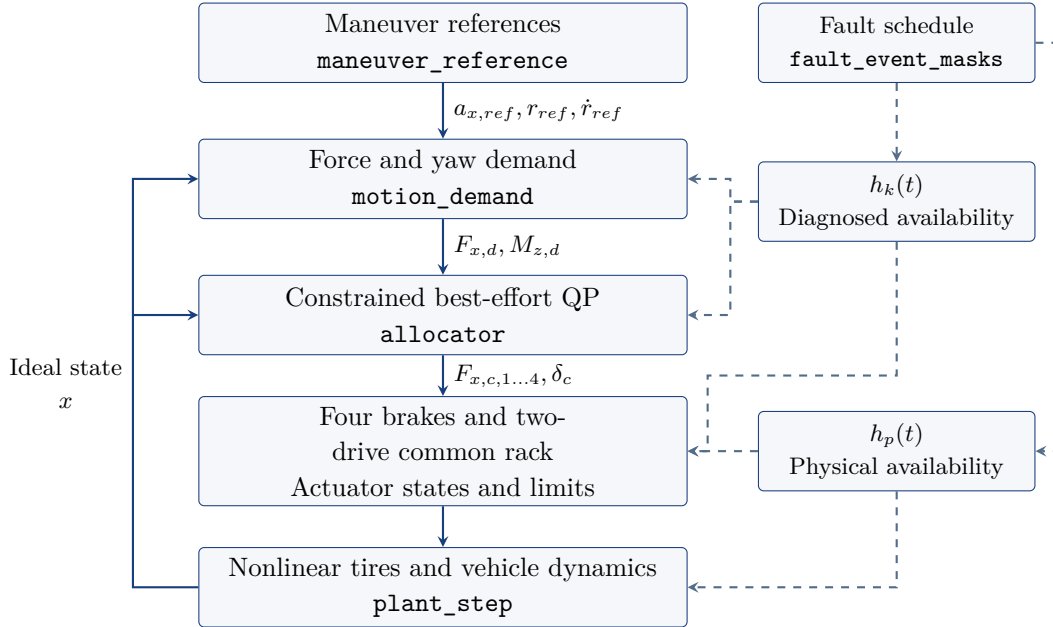
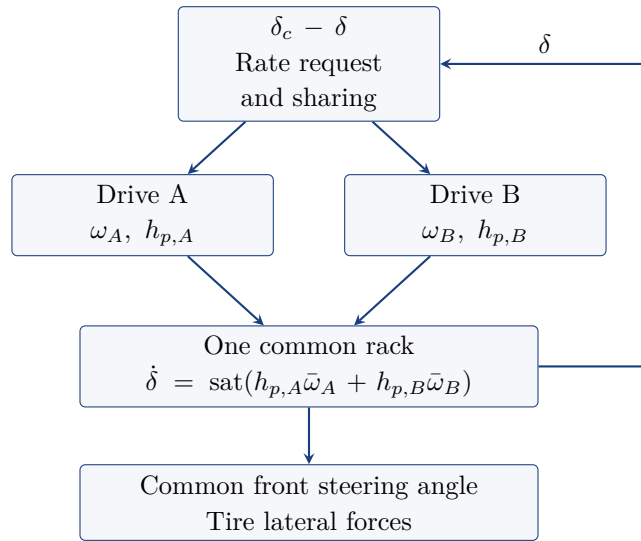


Figure 3.1 – Executed control and fault-injection chain. The controller also retains the previous command to impose rate limits.

The nominal controller period is 2 ms. The physical mask removes actuator capability at fault onset; the known mask changes only after the imposed diagnosis delay. The state feedback comes directly from the simulated plant. This architecture isolates diagnosis-latency sensitivity without claiming that a sensor-based detector or state estimator has been implemented.

### 3.5 Common-rack actuation and available redundancy



Two healthy drives:  $400^\circ/\text{s}$  aggregate limit    One healthy drive:  $240^\circ/\text{s}$   
 Both lost: held rack; brake asymmetry supplies backup yaw

Figure 3.2 – Functional steering model with two velocity-loop contributions and one mechanical degree of freedom.

Each motor loop has a 10 ms time constant; the outer angle-to-rate response uses 20 ms. The surviving drive receives the remaining request after diagnosis. Its assumed rate capability is 60% of the healthy pair's limit, while the allowed angle remains  $\pm 35^\circ$ . These are functional model parameters, not a measured torque-under-load specification. The four brake channels remain separately commanded. Their longitudinal tire forces contribute both to deceleration and to yaw according to wheel position. All channels share vehicle load transfer and tire adhesion. Thus the drawing of separate actuator channels does not imply independent power supplies, buses, sensors or safety controllers. Those dependencies are examined in the safety chapter and Appendix B.

## Chapter 4

# Vehicle and actuator modeling

---

### 4.1 State variables and equations of motion

The planar state is

$$x = [v_x, v_y, r, X, Y, \psi, \delta, F_{x,FL}, F_{x,FR}, F_{x,RL}, F_{x,RR}, \omega_A, \omega_B]^T.$$

For body-frame forces and yaw moment, the motion equations are

$$\begin{aligned} \dot{v}_x &= F_x/m + v_y r, & \dot{v}_y &= F_y/m - v_x r, & \dot{r} &= M_z/I_z, \\ \dot{X} &= v_x \cos \psi - v_y \sin \psi, & \dot{Y} &= v_x \sin \psi + v_y \cos \psi, & \dot{\psi} &= r. \end{aligned}$$

Front and rear slip angles are

$$\alpha_f = \delta - \text{atan2}(v_y + l_f r, v_x), \quad \alpha_r = -\text{atan2}(v_y - l_r r, v_x).$$

A simplified Magic Formula preserves the sourced small-slip axle stiffness. Each wheel's remaining lateral capacity is  $\sqrt{\max((\mu F_{z,i})^2 - F_{x,i}^2, 0)}$ . The axle peak is the sum of these capacities, and lateral force is distributed by capacity share. Front forces are rotated into the body frame before summing force and moment.

$$F_y = D \sin[C \arctan\{B\alpha - E(B\alpha - \arctan B\alpha)\}], \quad B = \frac{C_\alpha}{CD}.$$

Quasi-static load transfer uses an estimated longitudinal acceleration from actuator force and  $a_y \simeq v_x r$ . Negative raw wheel loads trigger a check even though force evaluation clips them to zero. Brake states use a 20 ms lag limited to 40 kN/s. A 20 ms rack outer loop distributes requested velocity to two 10 ms motor loops. Classical RK4 integrates between allocation samples. Validity is restricted to  $v_x > 0.5$  m/s and  $|\beta| \leq 0.15$  rad, with  $\beta = \text{atan2}(v_y, v_x)$ . The model omits wheel-speed dynamics, slip-ratio control, rack inertia, road-load torque, suspension transients and sensor dynamics. Passing numerical tests is model verification, not experimental validation of a real vehicle.

### 4.2 Load transfer and combined-force coupling

Longitudinal load transfer changes the front/rear distribution according to the estimated acceleration. Lateral transfer redistributes each axle's load between its left and right wheels. The model evaluates raw loads before clipping, allowing lift-off conditions to be detected numerically.

$$\Delta F_{z,x} = \frac{mh_{CG}a_x}{L}, \quad F_{zf} = mg \frac{l_r}{L} - \Delta F_{z,x}, \quad F_{zr} = mg \frac{l_f}{L} + \Delta F_{z,x}.$$

For each wheel, longitudinal and lateral forces share the same friction-circle budget. Consequently, using nearly all available tire force for braking leaves little lateral support. This is central to the degraded-control design: yaw balance alone is not sufficient to preserve sideslip behavior.

### 4.3 Parameter provenance

REF-2026-02 is a literature-based reference vehicle, not an identified target vehicle. S denotes sourced, C calculated, A assumed and R derived from the project requirement interpretation. Mixed provenance is intentional and must accompany the results.

| Parameter                             | Value                             | Provenance                           |
|---------------------------------------|-----------------------------------|--------------------------------------|
| $m, I_z$                              | 1590 kg; 2059.2 kg m <sup>2</sup> | S, Wang et al. [7]                   |
| $l_f, l_r$                            | 1.05 m; 1.61 m                    | S [7]                                |
| $L, w_f$                              | 2.66 m; 0.6053                    | C: $l_f + l_r, l_r/L$                |
| $t_w$                                 | 1.53 m                            | A: width used as track surrogate [8] |
| $h_{CG}$                              | 0.637 m                           | S surrogate [8]                      |
| $g, \mu$                              | 9.80665 m/s <sup>2</sup> ; 0.90   | S [9]; A                             |
| $C_f = C_r$                           | 66000 N/rad                       | C: $2 \times 33000$ [7]              |
| $C, E$                                | 1.9; 0.97                         | A                                    |
| $D, B$                                | Capacity-dependent                | C, previous section                  |
| $\delta_{\max}$                       | 35°                               | A                                    |
| Rack / drive rate                     | 400°/s; 240°/s                    | R/A; A                               |
| Brake force rate                      | 40000 N/s                         | R/A                                  |
| $\tau_{Fx}, \tau_\delta, \tau_\omega$ | 20, 20, 10 ms                     | A                                    |
| Yaw servo                             | 0.25 s / 0.15 s (DBBS)            | A                                    |
| Controller step                       | 2 ms                              | R                                    |
| Initial speed                         | 60 km/h                           | R                                    |
| Force reserve                         | 2% / 8% rear after front loss     | A                                    |
| Objective scales                      | 5000 N; 1000 N m                  | A                                    |
| Effort weight / trust angle           | $10^{-6}$ ; 0.5°                  | A                                    |

The 1.53 m value in [8] is vehicle width, not a measured track. Transferring it into the track parameter is explicitly an assumption. Geometry, tire shape, friction and actuator dynamics therefore limit representativeness. The 60% surviving steering rate is built into the functional model; the bench checks implementation consistency rather than discovering a measured capacity. One unsaturated 20 ms lag reaches 91.8% in 50 ms. This calculation does not prove the cascaded steering recovery requirement. Measured vehicle data and parameter sweeps would be needed before transferring results to a particular vehicle.

#### 4.4 Verification before allocation

Plant verification preceded closed-loop allocation. Straight braking checks longitudinal deceleration and near-zero yaw under symmetric commands. Step steering checks the sign and qualitative transient of yaw and lateral response. High-force cases check tire saturation and positive loads. Separate common-rack benches exercise two healthy drives, each single-drive loss and complete drive loss.

The preserved Gate 2 campaign includes three plant-only cases and four steering cases. This hierarchy is important: a nominal closed-loop trajectory alone would not reveal whether an incorrect vehicle equation had been compensated accidentally by the controller. The tests establish internal model consistency at the selected conditions; physical vehicle correlation is a subsequent perspective.

# Allocation strategy and degraded-mode design

---

## 5.1 Constrained quadratic allocation

At each sample, the allocator approximates generalized output by  $\hat{y} = c + Bu$ , with desired  $y_d = [F_{x,d}, M_{z,d}]^T$ . The offset matches actual nonlinear forces at the current anchor. Brake columns use force geometry with frozen lateral forces and loads. A centered secant estimates steering yaw sensitivity. Steering is not credited with a longitudinal sensitivity in the QP, avoiding an identified saturation-induced steering/braking feedback problem. Actual longitudinal steering effects remain in the offset and plant.

$$\min_z \frac{1}{2} \|W(c + BSz - y_d)\|_2^2 + \frac{\rho}{2} \|z\|_2^2, \quad u = Sz,$$

$$W = \text{diag}(1/5000, 1/1000), \quad \rho = 10^{-6}.$$

$S$  scales the four forces and rack angle into dimensionless decisions. The strictly positive effort penalty regularizes the Hessian. The constraints intersect tire/actuator bounds, healthy command-rate bounds, fault masks and a local steering trust interval. Brake forces remain non-positive. Nominal integrated commands retain a 2% longitudinal reserve; the rear-brake policy becomes 8% after a diagnosed front-brake loss with both rear brakes healthy.

$$\begin{aligned} -\gamma_i \mu F_{z,i} \leq u_i \leq 0, & \quad |u_i - u_{i,\text{prev}}| \leq \dot{F}_{\text{max}} \Delta t, \\ |\delta_c| \leq 35^\circ, & \quad |\delta_c - \delta_{c,\text{prev}}| \leq \dot{\delta}_{\text{avail}} \Delta t. \end{aligned}$$

A failed brake has zero force authority. With both steering drives lost, the rack command equals the held actual angle. If physical bounds and rate bounds have an empty intersection, physical bounds take priority and the healthy-channel override is explicitly logged as a validation issue. No such healthy override occurs in the cited accepted numerical runs. Demand infeasibility is different from solver failure. An infeasible request still yields the closest weighted bounded output. A failed QP instead clips the previous command into the current bounds and raises a warning. This fallback is not an independent safety controller. The finite-dimensional QP prediction is also not a global nonlinear reachable-set oracle.

## 5.2 Rear lateral reserve after front-brake loss

The final controller sets  $\gamma_i = 0.98$  in normal allocation. When exactly one front brake is diagnosed unavailable and both rear brakes remain healthy, rear channels use  $\gamma_i = 0.92$ . This leaves a larger theoretical lateral-force budget at the commanded longitudinal limit. The trigger

is diagnosed health, never a test-case identifier.

$$\sqrt{1 - 0.98^2} = 0.199, \quad \sqrt{1 - 0.92^2} = 0.392.$$

The reserve is a control choice, not a change to road friction or the tire equations. A design sweep over six rear-utilization values exposed the balance between residual braking and sideslip. The selected value retains more than 60% braking in the assessed plateau while substantially reducing peak sideslip. The margin is quantified in Chapter 6 rather than assumed to apply to all vehicles.

When a larger reserve is introduced, the controller approaches it at the permitted release rate. The reserve is softer than the physical bound. A genuinely empty intersection of physical and rate constraints is retained as a separate transition-model issue, discussed in the future-work chapter.

### 5.3 DBBS yaw-feedback scheduling

$$\tau_r = \begin{cases} 0.15 \text{ s}, & h_{k,A} = h_{k,B} = 0, \\ 0.25 \text{ s}, & \text{otherwise.} \end{cases}$$

Faster yaw feedback after complete diagnosed steering-drive loss improves pulse tracking and the resulting equal-distance path comparison. Single-drive loss and nominal operation retain the original gain. The external yaw reference is unchanged; this is not a hidden relaxation of the maneuver. A speed-scaled reference alternative was explored but not selected because it increased the measured path deviation.

### 5.4 Implementation and verification discipline

The controller is divided into reference generation, demand construction, constrained allocation and plant integration. Tests check default/explicit parameters, fault-mask behavior, rear-reserve activation, rate-compatible reserve introduction and diagnosis-based feedback scheduling. The final reference suite contains 45 passing MATLAB unit tests. A local Python audit adds eight tests for saved-data timing arithmetic. These are distinct suites and are not counted as 53 MATLAB tests.

## Simulation campaign and achieved performance

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### 6.1 Execution conditions and evidence

The reference campaign uses MATLAB Online R2026a Update 5 with Optimization Toolbox,  $\Delta t = 2$  ms,  $v_x(0) = 60$  km/h and  $\mu = 0.90$ . The final physical parameter set is REF-2026-02 and controller revision ALLOC-2026-04. Actual forces and moments are obtained from the nonlinear plant, not from the QP prediction. All figures below are plotted from preserved MATLAB CSV data; local plotting does not rerun the simulation.

### 6.2 Nominal maneuvers

G2-BRK is progressive straight braking at  $0.20g$ . G2-STR follows the linear bicycle yaw response associated with a  $1^\circ$  road-wheel input. G2-CMB combines  $0.12g$  braking with a  $0.06$  rad/s yaw demand. G2-YAW applies a windowed sinusoidal reference over 5 s, with coefficient  $0.08$  rad/s. G2-SAT requests  $1.2\mu g$  and then releases the brakes. The four tracking tests last 6 s; the capacity test lasts 4 s.

Pulse ramps use  $w(s) = 3s^2 - 2s^3$  on  $0 \leq s \leq 1$ , with zero endpoint slope and an analytic derivative for the yaw demand. The sinusoid uses a  $\sin^2$  window. These references are reproducible project maneuvers, not standardized vehicle tests.

Table 6.1 – Nominal maneuver results.

| Case   | RMSE( $F_x$ ), N | RMSE( $r$ ), rad/s | $ \beta _{max}$ , rad |
|--------|------------------|--------------------|-----------------------|
| G2-BRK | 44.65            | 0.00000            | 0.00000               |
| G2-STR | 20.62            | 0.00213            | 0.00511               |
| G2-CMB | 28.54            | 0.00179            | 0.00267               |
| G2-YAW | 6.27             | 0.00172            | 0.00326               |
| G2-SAT | 1541.32          | 0.00000            | 0.00000               |

The four tracking maneuvers meet their force and yaw RMSE criteria. The saturation maneuver deliberately requests more than the physical capacity, so it is evaluated on delivered capacity, bounded response and release behavior instead of force tracking. It reaches a saturated braking ratio of 0.98. The five saved nominal result summaries match the prior verified reference within the local audit tolerance.

### 6.3 Fault-scenario catalog

Table 6.2 – Thirty original fault scenarios.

| N  | Condition                                                 | Timing                               |
|----|-----------------------------------------------------------|--------------------------------------|
| 18 | FL/FR/RL/RR or steering A/B loss during combined maneuver | Onset 2 s; diagnosis 0/10/50 ms      |
| 6  | Complete steering-drive loss, positive DBBS pulse         | Onset 1 or 3 s; diagnosis 0/10/50 ms |
| 4  | Single brake loss during progressive braking              | Onset 2 s; immediate diagnosis       |
| 2  | Complete steering-drive loss, negative DBBS pulse         | Onset 1 or 3 s; diagnosis 10 ms      |

The 22 single-channel cases and eight complete-drive-loss backup cases are kept distinct. Four matched nominal references support comparison. Complete steering loss holds the actual rack position, giving centered and turning-rack cases. The pulse ramps to  $\pm 0.15$  rad/s, holds, then returns smoothly to zero; it is a project maneuver, not a standardized lane-change test.

### 6.4 Measurement conventions

Lateral deviation is measured in the nominal heading frame at equal cumulative travelled distance. This avoids confusing a speed difference with a spatial tracking error. The nominal record includes sufficient coasting distance to cover the faulted path without extrapolation. With nonzero sideslip, this is a heading-normal deviation, not the exact shortest geometric distance.

$$e_{lat}(s) = -(X_f(s) - X_n(s)) \sin \psi_n(s) + (Y_f(s) - Y_n(s)) \cos \psi_n(s).$$

Yaw RMSE and DBBS peak deceleration are evaluated after physical onset. The first 200 ms yaw-continuity diagnostic compares synchronized nominal and faulted signals. These definitions are retained across controller revisions so that improvements are attributable to the design rather than the measurement method.

## 6.5 DBBS performance

Table 6.3 – DBBS results with ALLOC-2026-04.

| Case            | $ e_{lat} _{max}, \text{ m}$ | RMSE( $r$ ) | $a_{max}/g$ |
|-----------------|------------------------------|-------------|-------------|
| DBBS-T1-D00     | 0.2945                       | 0.01146     | 0.2771      |
| DBBS-T3-D00     | 0.2617                       | 0.00830     | 0.2328      |
| DBBS-T1-D10     | 0.2945                       | 0.01146     | 0.2771      |
| DBBS-T3-D10     | 0.2617                       | 0.00830     | 0.2328      |
| DBBS-T1-D50     | 0.2945                       | 0.01146     | 0.2771      |
| DBBS-T3-D50     | 0.2616                       | 0.00830     | 0.2328      |
| DBBS-NEG-T1-D10 | 0.2945                       | 0.01146     | 0.2771      |
| DBBS-NEG-T3-D10 | 0.2617                       | 0.00830     | 0.2328      |

We observe negligible sensitivity to the tested 0/10/50 ms diagnosis delays because diagnosis precedes the pulse in the 1 s fault cases: even the latest flag arrives at 1.05 s, before the pulse starts at 1.5 s. The 3 s cases occur on a constant-reference plateau with the rack held at its existing angle. This finding is specific to these fault timings and the held-rack model; it does not establish general latency robustness during rapid transients. By contrast, Table 8.1 examines abrupt brake loss during saturated braking, which immediately disturbs the force balance.

All eight cases satisfy the three maneuver criteria. The worst path deviation is reduced from 0.5086 m in the initial design to 0.2945 m in the corrected design, approximately a 42% reduction. The maximum deceleration increases from 0.2577 g to 0.2771 g, illustrating the physical cost of better yaw tracking through differential braking. The comparison is between two revisions of the same architecture, not between unrelated controller families.

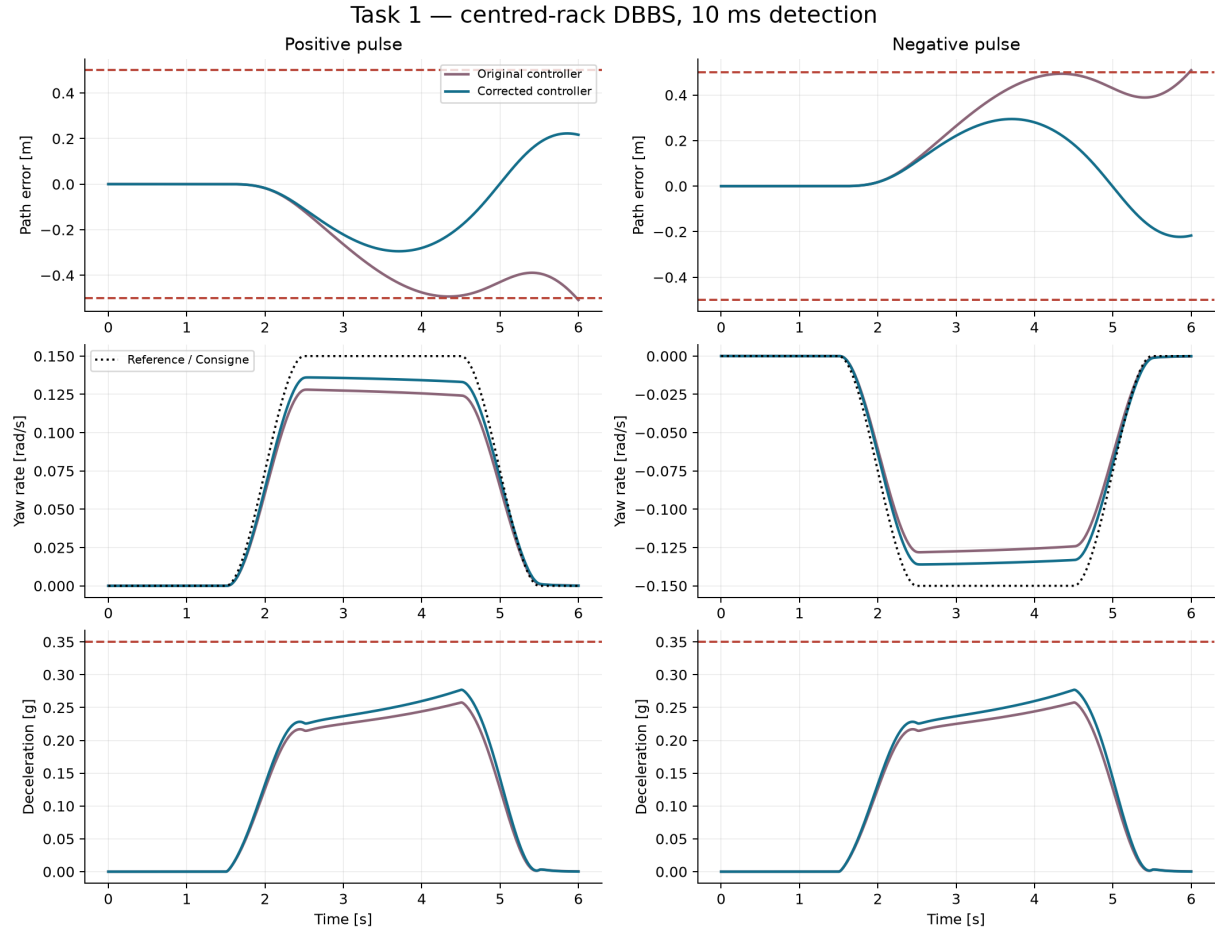


Figure 6.1 – Centred-rack DBBS before and after correction, both yaw directions, 10 ms diagnosis.

## 6.6 Residual braking and sideslip

The dynamic authority maneuver applies the channel loss before the heavy-braking demand. On the unchanged 1.5–1.9 s plateau, braking is normalized by  $\mu g$ , a nominal upper bound. This deliberately conservative denominator supports a retained-capability statement without assuming a measured maximum from a different maneuver.

Table 6.4 – Delivered braking authority in the evaluated plateau.

| Condition     | $a/(\mu g)$ | $ \beta _{max}$ , rad | $ M_z _{hold}$ , N m |
|---------------|-------------|-----------------------|----------------------|
| nominal       | 0.9800      | 0.00000               | 0.000                |
| brake_fl_loss | 0.6111      | 0.06051               | 10.023               |
| brake_fr_loss | 0.6111      | 0.06051               | 10.023               |
| brake_rl_loss | 0.8799      | 0.00723               | 2.663                |
| brake_rr_loss | 0.8799      | 0.00722               | 2.663                |

The front-loss cases retain 61.1147% braking, with peak sideslip about 0.0605 rad and hold

yaw moment about 10.0 N m. Rear-loss cases retain 87.9923%. Compared with the initial front-loss tuning, retained braking decreases by 2.5642 percentage points while peak sideslip decreases from 0.2416 to 0.0605 rad. This is a deliberate stability/capability compromise; the remaining margin above 60% is 1.1147 percentage points.

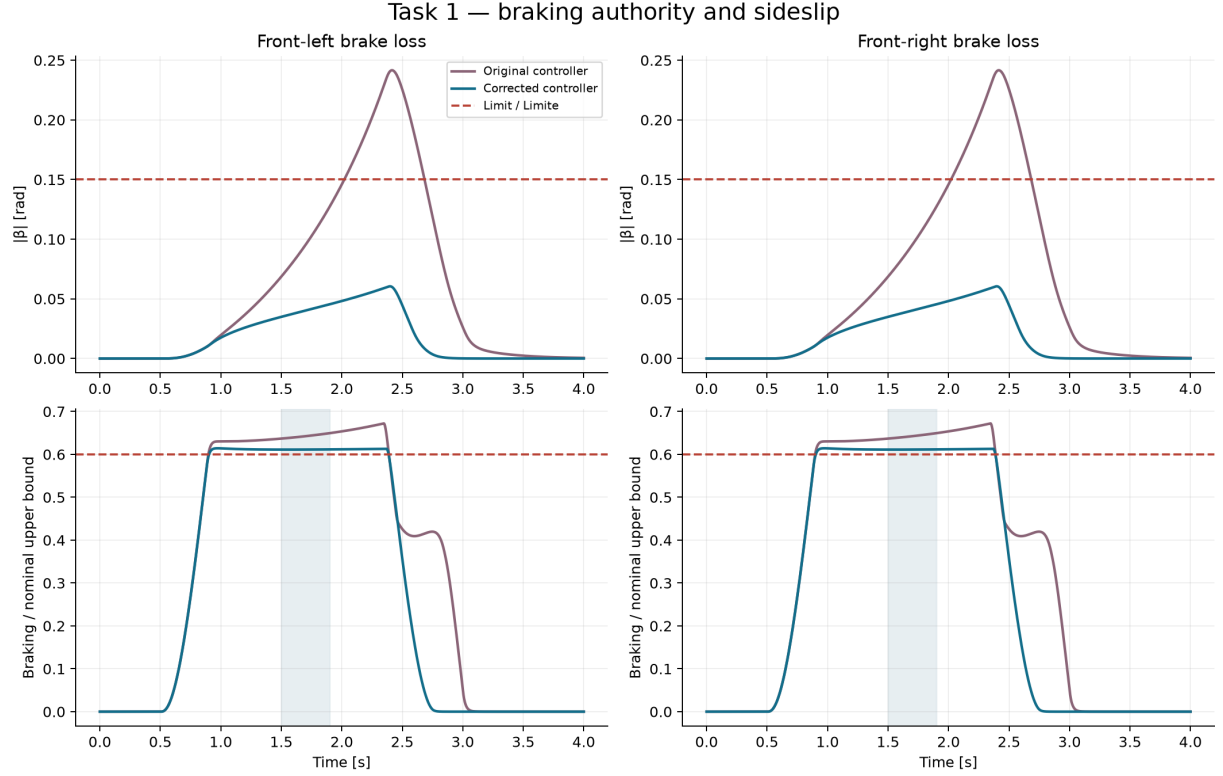


Figure 6.2 – Front-brake-loss authority: rear-grip reserve improves sideslip with a quantified braking trade-off.

## 6.7 Independent authority witnesses

Separate nonlinear searches construct physically feasible static witnesses independently of the allocator’s affine output. A force-aligned straight-path witness enforces zero yaw rate and moment and aligns resultant force with velocity. Its worst retained braking lower bound is 61.2636%. These witnesses establish existence at the assessed condition, not global optimality or guaranteed dynamic reachability.

$$F_{upper} = \mu mg, \quad M_{upper} = F_{upper} \max \left( \sqrt{l_f^2 + (t_w/2)^2}, \sqrt{l_r^2 + (t_w/2)^2} \right).$$

For centered DBBS at zero initial slip, the feasible yaw witness under a 0.35 g deceleration cap is approximately 4174.91 N m in either direction. Relative to the conservative nominal yaw upper bound, this gives 16.6900%, above the 15% project authority benchmark. The witness and the tracking maneuver answer different questions and are therefore reported separately.

## 6.8 Timing diagnostics and numerical refinement

The local audit independently recomputed all 30 original mask, affine-recovery and continuity diagnostics. The masks are applied in the diagnosis sample. All cases are already within the affine comparison band at that sample and remain there for the required dwell; the resulting zero entry delay must therefore not be interpreted as an instantaneous physical settling time. The maximum first-200-ms yaw difference is 0.013688 rad/s.

Table 6.5 – Targeted 2 ms versus 1 ms comparison.

| Case                  | $\Delta r_{max}$ , rad/s | $\Delta v_{max}$ , m/s | $\Delta e_{peak}$ , mm |
|-----------------------|--------------------------|------------------------|------------------------|
| DBBS +                | 0.000156                 | 0.002192               | 0.727                  |
| DBBS -                | 0.000156                 | 0.002106               | 0.731                  |
| Front-left authority  | 0.000589                 | 0.002861               | –                      |
| Front-right authority | 0.000589                 | 0.002860               | –                      |

The centred DBBS path result remains below 0.2953 m at 1 ms. Refinement supports numerical consistency of the selected cases. It does not replace parameter uncertainty analysis or physical vehicle testing.

## 6.9 Cost of diagnostic latency: oracle-gap analysis

Every fault run performs a second allocation at the same frozen state, with the same previous command and the same demand, but using the *physical* mask instead of the diagnosed one. This second solution is the oracle: the command a perfectly informed allocator would have issued at that instant. It is logged in each time series as `OracleFx_N`, `OracleMz_Nm` and the associated authority scales. Comparing the delivered nonlinear output with the oracle isolates the price of diagnostic latency from the price of the demand itself being infeasible.

$$e_y(t) = \left\| (y(t) - y_{best}(t)) \oslash y_{scale}(t) \right\|_{\infty}, \quad y = [F_x, M_z]^T.$$

We evaluate two windows. The latency window  $[t_p, t_f)$  is the interval during which the controller still treats a lost effector as available. The reconfiguration window  $[t_f, t_f + 100 \text{ ms}]$  follows the flag, when the command must move to its new operating point under the rate limits. The oracle remains an affine allocator prediction; the measured side of the comparison is the nonlinear plant output.

Table 6.6 – Normalized oracle gap, recomputed from the archived time series.

| Case family              | $e_y$ latency 10 ms | $e_y$ latency 50 ms | $e_y$ reconfig. | $e_y$ peak |
|--------------------------|---------------------|---------------------|-----------------|------------|
| Front-left brake loss    | 0.0241              | 0.0242              | 0.0597          | 0.0597     |
| Front-right brake loss   | 0.0258              | 0.0259              | 0.0620          | 0.0620     |
| Rear-left brake loss     | 0.0206              | 0.0207              | 0.0222          | 0.0222     |
| Rear-right brake loss    | 0.0214              | 0.0214              | 0.0223          | 0.0223     |
| Steering drive A loss    | 0.00004             | 0.00005             | 0.00005         | 0.0056     |
| Steering drive B loss    | 0.00004             | 0.00005             | 0.00005         | 0.0056     |
| DBBS, fault at 1 s       | 0.0000              | 0.0000              | 0.0000          | 0.0192     |
| DBBS, fault at 3 s       | 0.0000              | 0.0000              | 0.0002          | 0.0138     |
| Ramp braking, front loss | –                   | –                   | 0.0898          | 0.0898     |
| Ramp braking, rear loss  | –                   | –                   | 0.0383          | 0.0383     |

Three observations follow. First, the gap accumulated while the controller is misinformed never exceeds 0.0259 for a brake loss and is below  $10^{-4}$  for a steering-channel loss, because the surviving drive absorbs the missing velocity contribution without a change of operating point. Second, and less obviously, the reconfiguration window is the larger contributor for every brake case: the dominant transient cost is the rate-limited move to the new command, not the delay in learning about the fault. Third, extending the imposed delay from 10 ms to 50 ms changes the peak gap by less than  $10^{-3}$  in normalized units in every catalog case, which is the quantitative form of the insensitivity noted in section 6.5.

For the DBBS cases the peak gap does not occur at the fault at all, but during the yaw pulse itself, reaching 0.0192 for the centred rack and 0.0138 for the held cornering rack. This is consistent with the held-rack model: the fault removes no force at the instant it occurs, and the allocator only has to work when lateral demand arrives. Every value in the table stays inside the 10% normalized band of REQ-02, but the comparison target remains the affine oracle, so this supports rather than establishes the recovery clause.

## 6.10 Sampled envelope assessment

The earlier envelope assessment tested 300 target points at 60 frozen states. It obtained 240 feasible witnesses and 236 qualifying 50 ms endpoint constructions. Reuse with the corrected controller was checked rather than assumed: 44 single-channel sampled states have matching inputs, allowing 220 witnesses and 216 endpoint results to transfer. The 16 DBBS states changed with the controller and their earlier probe results are not automatically reused.

This sampled evidence is useful engineering information. It is not a continuous envelope proof: successful box corners do not establish a complete nonlinear reachable set, and a 50 ms endpoint does not establish a 100 ms dwell. Extending this evidence is a clearly identified future perspective.

# Safety rationale and engineering synthesis

## 7.1 Functional safety analysis

The safety work organizes the functional architecture into braking, lateral-control and unintended-actuation concerns. FMEA links component-level causes to vehicle-level effects and compensating actions. The scope is a qualitative engineering rationale, not quantified hardware reliability or an assigned ASIL obtained from a full hazard analysis. ISO 26262-9 provides the framework for ASIL-oriented decomposition, dependent-failure and safety analyses [10].

Table 7.1 – Condensed FMEA and development perspectives.

| Element / mode                   | Effect and current treatment                        | Continuation                           |
|----------------------------------|-----------------------------------------------------|----------------------------------------|
| One brake lost                   | Reduced/asymmetric braking; mask and redistribution | High-saturation transition dynamics    |
| One steering drive lost          | Reduced rack rate; surviving drive remains active   | Road-load torque and electrical sizing |
| Both drives unavailable          | Held rack; DBBS yaw compensation                    | Wider speed/friction envelope          |
| Dedicated power / communication  | Represented by equivalent isolated channel loss     | Actual isolation and timeout behavior  |
| Shared ECU / sensing             | Shared dependency; ideal input assumptions          | Independent monitor and estimation     |
| Mechanical jam / stuck actuation | Different physical effect from a released channel   | Dedicated models and mitigations       |

The complete F01–F18 analysis, including separate actuator, power and communication causes and the D01–D08 dependency register, is provided in Appendix B. The table above is its reading guide, not the full safety register.

## 7.2 Candidate hazard analysis and safety goals

The report refers to an ASIL D target without deriving it. We therefore record the candidate hazard classification that the target implies, so that each requirement of Chapter 3 can be attached to a safety goal rather than appearing as a free-standing number. Severity, exposure and controllability are assigned qualitatively, at the vehicle level, for a passenger car in normal road use. This is a candidate classification prepared to structure the analysis; a certified HARA requires an agreed item definition, operational situation catalog and controllability study that

this project did not have.

Table 7.2 – Candidate hazards, classification and derived safety goals.

| ID | Vehicle-level hazard                                         | S/E/C    | ASIL   | Safety goal |
|----|--------------------------------------------------------------|----------|--------|-------------|
| H1 | Loss of requested deceleration during a braking request      | S3/E4/C3 | ASIL D | SG-01       |
| H2 | Unintended yaw moment during a straight-line braking request | S3/E4/C3 | ASIL D | SG-02       |
| H3 | Loss of commanded lateral control                            | S3/E4/C3 | ASIL D | SG-03       |
| H4 | Unintended or opposing rack motion                           | S3/E4/C3 | ASIL D | SG-04       |
| H5 | Unintended brake force at one corner                         | S3/E4/C3 | ASIL D | SG-05       |

Table 7.3 – Safety goals, implemented mitigation and evidence status.

| ID    | Safety goal                                                                                                 | Mitigation in this demonstrator                                     | Status                       |
|-------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------|
| SG-01 | Retain at least 60% of nominal straight-line deceleration after any single effector loss                    | Masked QP redistribution across the surviving corners               | Tested, section 6.6          |
| SG-02 | Limit unintended yaw moment during a straight braking request to 5% of nominal yaw authority                | Explicit $M_z$ term in the QP objective with fixed priority scaling | Tested, section 6.6          |
| SG-03 | After loss of commanded steering, provide at least 15% of nominal yaw authority with bounded path deviation | DBBS with scheduled yaw feedback and held rack                      | Tested, sections 6.5 and 6.7 |
| SG-04 | Prevent unintended or opposing rack torque                                                                  | None. The velocity-sharing model represents loss, not opposition    | Not addressed; F18           |
| SG-05 | Detect and remove unintended corner brake force                                                             | None. Fault injection removes force, it never adds force            | Not addressed; F17           |

Recording SG-04 and SG-05 with no mitigation is deliberate. Both correspond to unintended-actuation modes already listed as F17 and F18, and both require an independent monitor rather than a reallocation policy: an allocator that can only command the effectors it believes in cannot remove an effector output it did not request. This is a structural limit of the present architecture, not a tuning gap.

### 7.3 Fault-tolerant time interval budget

The imposed diagnosis delays of section 3.2, the 10 ms mask benchmark of section 3.3, the recovery diagnostics of section 6.8 and the oracle-gap results of section 6.9 all measure parts

of one quantity: the interval between physical fault onset and restored controlled behavior. Collecting them into a single budget makes the safety argument checkable and shows which term dominates.

$$\text{FTTI} \geq t_{\text{detect}} + t_{\text{react}} + t_{\text{settle}},$$

Table 7.4 – Budget contributions in the present demonstrator.

| Contribution                          | Value                  | Basis and status                                                               |
|---------------------------------------|------------------------|--------------------------------------------------------------------------------|
| Fault detection $t_d$                 | 0 / 10 / 50 ms         | Imposed sensitivity cases; no sensor-based detector is implemented             |
| Mask application                      | $\leq 2$ ms            | One allocation period; sampled on the control grid                             |
| Brake command slew, full corner range | $\approx 106$ ms front | 4246.6 N front corner capacity at 40 kN/s; 2770.1 N rear gives $\approx 69$ ms |
| Brake force lag                       | 20 ms ( $1\tau$ )      | First-order model; 91.8% of an unsaturated step in 50 ms                       |
| Rack slew, full range                 | $\approx 175$ / 292 ms | $\pm 35^\circ$ at 400°/s healthy pair, 240°/s single drive                     |
| Settling within the 10% band          | peak $e_y \leq 0.0898$ | Measured, section 6.9; affine oracle reference                                 |

No FTTI target is assigned, because no agreed hazard analysis supplied one. The budget is nevertheless already decisive for one clause. In 50 ms the brake rate limit permits a command change of 2000 N, which is 47.1% of a front corner’s friction-limited range and 72.2% of a rear corner’s. The 50 ms recovery clause of REQ-02 is therefore physically attainable only when the required reconfiguration is smaller than those fractions; for a larger required change no allocation policy can satisfy it, and the constraint is the actuator rate, not the controller. Stating this explicitly is more useful than reporting a diagnostic band entry of zero, and it identifies the brake force-rate limit as the parameter to revisit first if the clause is retained.

## 7.4 Degradation states and safe state

The demonstrator switches behavior on the diagnosed mask, but it never names the resulting vehicle modes. Naming them is required before any warning and degradation concept can be written, and it clarifies what the measured results correspond to.

Table 7.5 – Modes implied by the implemented mask logic.

| Mode                | Entry condition            | Implemented behavior                                                        |
|---------------------|----------------------------|-----------------------------------------------------------------------------|
| M0 nominal          | All six channels healthy   | Full allocation, 2% longitudinal reserve, 0.25 s yaw feedback               |
| M1 reduced braking  | One brake diagnosed lost   | Redistribution; 8% rear reserve if the loss is at the front                 |
| M2 reduced steering | One drive diagnosed lost   | Surviving drive takes the remaining rate demand                             |
| M3 DBBS             | Both drives diagnosed lost | Rack held at its current angle; 0.15 s yaw feedback; yaw by brake asymmetry |
| M4 safe state       | Not implemented            | No minimal risk maneuver, no controlled stop, no driver handover is modeled |

M4 is the honest entry. Every measured result in Chapter 6 is a transition into M1, M2 or M3 followed by continued driving; the demonstrator has no notion of terminating the degraded condition. A production concept would require the safe state, its entry criteria, the driver warning and the maximum permissible dwell in M3, none of which are within this scope.

## 7.5 Qualitative fault tree and independence

Let  $C$  denote an unmitigated common cause,  $I_S/I_B$  loss of command or control,  $J$  an uncontrollable rack state,  $A/B$  loss of the steering branches,  $D$  insufficient DBBS authority and  $R_B$  insufficient residual braking. A useful qualitative abstraction is

$$T_S = C \vee I_S \vee J \vee (A \wedge B \wedge D), \quad T_B = C \vee I_B \vee R_B.$$

Availability of healthy brakes alone does not guarantee sufficient backup steering; the relevant force and moment must be achievable in the current state. Similarly, two steering motors do not demonstrate independent sensing, computation, power and communication. A candidate ASIL D decomposition into B(D)+B(D) can be discussed conceptually, but actual branch independence must be justified at system level [11].

The shared dependencies identified are power and ground, communications, allocator software, state/request sensing, the common rack, road friction, diagnosis timing and physical installation. Recording them makes the architecture traceable and provides a concrete basis for future industrialization. No hardware certification is claimed by the simulation results.

## 7.6 Achievement of our project objectives

Table 7.6 – Summary of our contributions.

| Objective                  | Completed contribution                                                                               |
|----------------------------|------------------------------------------------------------------------------------------------------|
| Integrated modeling        | 13-state nonlinear plant, four brakes, two-drive common rack                                         |
| Functional allocation      | Constrained QP, diagnosed masks, bounded best effort                                                 |
| Degraded performance       | Quantified catalog results, corrected DBBS and front-loss tuning                                     |
| Engineering verification   | Unit tests, nominal/fault campaigns, independent local audits and refinement                         |
| Traceability and reporting | Source/evidence preservation, safety rationale, requirement matrix of Appendix C, FR/EN final report |

The academic premise is fulfilled through a functioning demonstrator whose retained capabilities are measured and explainable. The project does not equate this outcome with compliance with every initial all-condition requirement. The next chapter records the boundary and the technical perspectives that follow from the exploratory evidence.

## Scope limits and future perspectives

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### 8.1 Abrupt faults under fully saturated braking

Six exploratory cases introduce a front-brake loss during maximum braking, rather than before the demand. The requested deceleration is 120% of the global nominal friction bound, outside the current completion scope’s maneuver guarantees. These cases identify an actuator/contact-force modeling issue to address in a future system revision.

At the recorded onset, the healthy front command is approximately  $-5614.37$  N and the new physical lower bound is  $-5123.93$  N. Reaching that bound requires a  $490.44$  N release, while the 2 ms command-rate allowance is only 80 N. The intervals do not intersect at this state. The current implementation prioritizes the physical bound and records one rate override per exploratory case; this should not be interpreted as a physically rate-compliant actuator transition.

Table 8.1 – Exploratory transition indicators retained for future work.

| Diagnosis | Min. ratio | $\Delta r_{200ms}$ | Joint plateau dwell after diagnosis   |
|-----------|------------|--------------------|---------------------------------------|
| 0 ms      | 0.5025     | 0.05062            | 148 ms                                |
| 10 ms     | 0.5059     | 0.06014            | 182 ms                                |
| 50 ms     | 0.4959     | 0.10500            | No complete 100 ms dwell before 1.9 s |

The values are mirrored across front-left/front-right losses. The onset-inclusive yaw-moment peak is about  $3919.81$  N m, while peak sideslip stays below  $0.0604$  rad. The dwell combines braking at least 60% and the yaw plateau limit; it is descriptive and is not the independent recovery criterion. We identify these transitions as future development: the present results do not demonstrate rate compliance throughout the entire transition.

A useful continuation is to separate rate-limited actuator demand/state from instantaneous tire-contact saturation and, where justified, add suspension/load-transfer dynamics. The revised semantics would then be checked on a small targeted regression before extending the scenario set. Excluding a condition from the present report’s completion scope does not imply that the implemented controller enforces an operational exclusion automatically.

### 8.2 Broader envelope, recovery and robustness

The current timing result is an affine-reference diagnostic. A future independent nonlinear reachable-output calculation should distinguish mask activation, physical recovery, diagnosis latency and sustained tracking. The 15% reserve interpretation should be evaluated along tra-

jectories, with particular care for DBBS force/yaw coupling. At zero slip with a centered rack, yaw requires braking, so independent maxima do not define a rectangular feasible region.

The evaluated vehicle uses fixed mass, inertia, geometry, tire parameters and friction. Sensitivity to these assumptions, estimation errors and actuator time constants would strengthen transferability. The 1.1147 percentage-point braking margin particularly motivates uncertainty analysis before a vehicle-specific claim. A simple rule-based or pseudoinverse controller would provide an additional comparative baseline; the present before/after results concern only revisions of the QP-based system.

### 8.3 Industrialization perspectives

Further work can address wheel rotational dynamics and slip control, rack torque and inertia, measured motor capability, realistic sensors and diagnostic logic, independent monitoring, hardware-in-the-loop execution and real-time solver timing. Standardized maneuvers such as a double lane change or split-friction braking and a combined steering/brake-circuit loss form natural extensions. Their procedures should be specified before execution rather than relabelling the present yaw pulse as a standardized test.

# General conclusion

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Through this project, we deliver an integrated BbW/SbW simulation demonstrator that coordinates redundant functional actuators and adapts allocation to diagnosed channel availability. The nonlinear plant, constrained best-effort QP, fault semantics and reproducible evaluation chain form a complete academic implementation. The work connects vehicle dynamics, optimization, mechatronic architecture and safety reasoning within one coherent system.

The main quantitative outcomes are retained front-loss braking above 61%, controlled sideslip around 0.0605 rad in the authority tests, and DBBS path deviation below 0.295 m in the original maneuver catalog. The design improvements expose useful engineering trade-offs: retaining rear lateral support slightly reduces braking utilization, while faster backup yaw tracking requires more deceleration. Targeted refinement and independent saved-data checks reinforce the credibility of these findings.

We have achieved our project objective by demonstrating useful residual braking and lateral-control behavior after selected actuator losses, within the documented model and maneuver conditions. Our transparent, reusable simulation implementation provides a basis for further development. We identify a wider operating domain and industrial validation as the next technical stages, supported by the exploratory data preserved throughout our work.

## Appendix A

# Reproducibility and evidence organization

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### A.1 Reference software and source snapshot

The reference simulation uses MATLAB R2026a Update 5 and Optimization Toolbox. The main scripted campaigns do not require the legacy Simulink prototype. Controller sources executed in the final campaign are archived with the evidence, so a later worktree is not confused with the tested revision. The current report package includes a source snapshot and a SHA-256 manifest.

```
results = run_task1_validation();
```

This optional command creates a new timestamped output folder; an explicitly existing destination is rejected. It is documented for reproducibility, not executed during final report preparation. Reading results and rebuilding this report do not consume MATLAB Online simulation time.

### A.2 Private evidence index

Table A.1 – Evidence hierarchy.

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| Baseline           | Gate 2 nominal/plant/steering and initial Gate 3 campaigns      |
| Authority studies  | Nonlinear witnesses, force-aligned braking, sampled envelope    |
| Corrected campaign | TASK1-CTRL-01 archive; SHA-256 in package manifest              |
| Local assessment   | TASK2-LOCAL-01 audit, revision 2; JSON and seven CSV tables     |
| Final deliverable  | FR/EN LaTeX sources and PDFs, source snapshot, package manifest |

We verified the corrected campaign archive’s SHA-256 against the cloud copy. We preserve provenance through source matching and saved-data audits, and retain the original campaign status fields unchanged. Our public code repository excludes reports, source CDC documents and raw evidence archives.

### A.3 Local checks and report rebuilding

```
python validation/test_audit_task2.py
python validation/audit_task2.py validation/results/new_audit
tectonic final_report.en.tex
tectonic final_report.fr.tex
```

The audit requires the private saved evidence at its documented locations and only reads it. The report sources use the supplied logos, project-generated plots and retained technical sources. The final output consists of two report PDFs; defense slides and scripts are intentionally not generated.

## Appendix B

# Full FMEA and dependency register

We retain the complete F01–F18 register established during the safety work. F01–F06 each distinguish actuator (A), dedicated power-path (P) and dedicated communication-path (C) causes. These six families therefore contain 18 causal entries; with F07–F18, the register contains 30 coded entries. This count describes the analysis, not 30 independently validated physical fault mechanisms. The tables separate the modeled consequence from proposed detection and future implementation evidence. No numerical severity, occurrence, diagnostic coverage or risk-priority number is assigned without the necessary vehicle and hardware data. “Equivalent mask” means that an isolated fail-silent loss can be represented by the same unavailable channel; it does not establish electrical or communication independence.

### B.1 Brake and steering channel losses

For A entries, command/response plausibility is a candidate detection method; P entries call for local supply monitoring and isolation; C entries call for freshness, integrity and timeout monitoring. These mechanisms are proposed measures. In the tested model, diagnosis is imposed by the fault schedule.

Table B.1: Detailed channel-loss FMEA.

| ID   | Mode / cause                                   | Effect / response                                                                   | Evidence / continuation                                                                        |
|------|------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| F01A | Front-left brake unavailable                   | Reduced deceleration and asymmetric yaw; remove the channel and redistribute force. | G3-FL-D00/D10/D50, G3-BRK-FL and authority tests. Front-loss plateau retains 61.1147% braking. |
| F01P | Dedicated supply lost: front-left brake        | Same consequence only if isolated and fail-silent.                                  | Equivalent FL mask; physical isolation and supply diagnostics require hardware evidence.       |
| F01C | Dedicated communication lost: front-left brake | Same consequence only if timeout produces a defined unavailable state.              | Equivalent FL mask; actual network, message corruption and timeout behavior are not modeled.   |
| F02A | Front-right brake unavailable                  | Reduced deceleration and asymmetric yaw; remove the channel and redistribute force. | G3-FR-D00/D10/D50, G3-BRK-FR and authority tests. Front-loss plateau retains 61.1147% braking. |
| F02P | Dedicated supply lost: front-right brake       | Same consequence only if isolated and fail-silent.                                  | Equivalent FR mask; physical isolation and supply diagnostics require hardware evidence.       |

Continued on next page

| ID   | Mode / cause                                    | Effect / response                                                                   | Evidence / continuation                                                                                                                 |
|------|-------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| F02C | Dedicated communication lost: front-right brake | Same consequence only if timeout produces a defined unavailable state.              | Equivalent FR mask; actual network, message corruption and timeout behavior are not modeled.                                            |
| F03A | Rear-left brake unavailable                     | Reduced deceleration and asymmetric yaw; remove the channel and redistribute force. | G3-RL-D00/D10/D50, G3-BRK-RL and authority tests. Rear-loss plateau retains 87.9923% braking.                                           |
| F03P | Dedicated supply lost: rear-left brake          | Same consequence only if isolated and fail-silent.                                  | Equivalent RL mask; physical isolation and supply diagnostics require hardware evidence.                                                |
| F03C | Dedicated communication lost: rear-left brake   | Same consequence only if timeout produces a defined unavailable state.              | Equivalent RL mask; actual network, message corruption and timeout behavior are not modeled.                                            |
| F04A | Rear-right brake unavailable                    | Reduced deceleration and asymmetric yaw; remove the channel and redistribute force. | G3-RR-D00/D10/D50, G3-BRK-RR and authority tests. Rear-loss plateau retains 87.9923% braking.                                           |
| F04P | Dedicated supply lost: rear-right brake         | Same consequence only if isolated and fail-silent.                                  | Equivalent RR mask; physical isolation and supply diagnostics require hardware evidence.                                                |
| F04C | Dedicated communication lost: rear-right brake  | Same consequence only if timeout produces a defined unavailable state.              | Equivalent RR mask; actual network, message corruption and timeout behavior are not modeled.                                            |
| F05A | Steering drive a unavailable                    | Reduced rack slew rate; surviving drive takes the request after diagnosis.          | G3-SA-D00/D10/D50 and rack benches. Single-drive angle range is retained in the functional model; loaded torque remains to be assessed. |
| F05P | Dedicated supply lost: steering drive A         | Same consequence only if isolated and fail-silent.                                  | Equivalent SA mask; physical isolation and supply diagnostics require hardware evidence.                                                |
| F05C | Dedicated communication lost: steering drive A  | Same consequence only if timeout produces a defined unavailable state.              | Equivalent SA mask; actual network, message corruption and timeout behavior are not modeled.                                            |
| F06A | Steering drive b unavailable                    | Reduced rack slew rate; surviving drive takes the request after diagnosis.          | G3-SB-D00/D10/D50 and rack benches. Single-drive angle range is retained in the functional model; loaded torque remains to be assessed. |
| F06P | Dedicated supply lost: steering drive B         | Same consequence only if isolated and fail-silent.                                  | Equivalent SB mask; physical isolation and supply diagnostics require hardware evidence.                                                |
| F06C | Dedicated communication lost: steering drive B  | Same consequence only if timeout produces a defined unavailable state.              | Equivalent SB mask; actual network, message corruption and timeout behavior are not modeled.                                            |

## B.2 Shared, sensing and unintended-actuation modes

Table B.2: Detailed shared and non-loss failure modes.

| ID  | Mode / cause                                       | Effect / response                                                | Evidence / continuation                                                                                                                     |
|-----|----------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| F07 | Both steering drives lost                          | Held rack; lateral control depends on DBBS authority.            | Eight corrected DBBS cases meet their path/yaw/deceleration criteria. Common cause versus dual loss is not resolved by this test.           |
| F08 | Mechanical rack jam or free motion                 | Uncommanded steering state may exceed backup capability.         | Mechanical analysis, jam/free-rack models and bounded backup envelope are future work. Held-drive loss is not a general jam test.           |
| F09 | Steering request lost, frozen or corrupt           | The intended lateral motion may be unknown or incorrect.         | Candidate redundant input sensing and plausibility. No request fault is represented by an actuator mask.                                    |
| F10 | Brake request lost, frozen or corrupt              | Missing or unwanted deceleration demand.                         | Candidate independent request checks and defined fallback. No substitute driver intent is inferred in the model.                            |
| F11 | Rack-angle measurement bias or loss                | Incorrect servo error and allocation effectiveness.              | Ideal angle feedback is used. Independent sensing or qualified reconstruction requires dedicated verification.                              |
| F12 | Yaw or speed measurement bias or loss              | Incorrect yaw feedback, tire state or operating-domain estimate. | Ideal state feedback is used. Sensor plausibility and fallback estimation remain implementation perspectives.                               |
| F13 | Brake feedback fault at any corner                 | False isolation or delayed recognition of force loss.            | Four sensor instances require per-corner supervision. Sensor dynamics and diagnostic coverage are not simulated.                            |
| F14 | Allocator crash, overrun or plausible wrong output | Both force and yaw requests can be affected.                     | Bounded solver fallback is implemented. Watchdog, independent command monitor and qualified backup controller are separate future measures. |
| F15 | Late, missed or false diagnosis                    | Unavailable actuator retained, or healthy authority removed.     | 0/10/50 ms imposed delay sensitivity is tested. Missed, false and intermittent flags need additional cases and detector design.             |
| F16 | Common supply, ground or communication fault       | Correlated loss or corruption across domains.                    | Separate logical flags do not establish independence. Isolation schematics, common-cause analysis and hardware tests are required.          |
| F17 | Stuck-on or unintended brake force                 | Persistent asymmetric yaw or unwanted braking.                   | Candidate independent force supervision and disconnect path. Force-removal tests do not cover retained force.                               |

Continued on next page

| ID  | Mode / cause                                 | Effect / response                                        | Evidence / continuation                                                                                                                |
|-----|----------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| F18 | Unintended or opposing steering-drive torque | Rack motion can oppose the healthy drive or the command. | Torque supervision and disconnection require a loaded mechanical model. Velocity sharing does not establish opposing-torque tolerance. |

### B.3 Dependent-failure review

We use the following register to distinguish functional channel redundancy from implementation independence. These dependencies explain why a B(D)+B(D) decomposition remains a candidate system concept rather than an attribute that can be assigned to the two simulated motors alone.

Table B.3 – Shared dependencies and evidence needed for implementation.

| ID  | Dependency            | Evidence to develop                                                   |
|-----|-----------------------|-----------------------------------------------------------------------|
| D01 | Power and ground      | Supply routes, protection, isolation and common-failure analysis      |
| D02 | Communication         | Bus routes, integrity, freshness and timeout behavior                 |
| D03 | Shared computation    | Independent monitoring, execution timing and software-fault treatment |
| D04 | Inputs and sensors    | Sensor redundancy, diagnosis and fallback estimation                  |
| D05 | Common rack           | Jam, free motion, road load and motor torque capability               |
| D06 | Tire adhesion         | Friction/load sensitivity and simultaneous force/yaw envelope         |
| D07 | Diagnosis and timing  | Physical-onset timing budget, false/missed flags and recovery         |
| D08 | Physical installation | Thermal, electromagnetic and damage-related common causes             |

For the qualitative fault tree, the steering top event has minimal cut sets  $\{C\}$ ,  $\{I_S\}$ ,  $\{J\}$  and  $\{A, B, D\}$ ; braking has  $\{C\}$ ,  $\{I_B\}$  and  $\{R_B\}$ . A common cause appears once and is not treated as two independent events. These logical sets identify where an implementation needs protection; they do not supply failure probabilities.

## Requirements traceability

This matrix links each numerical clause of the engineering baseline to the implementation that addresses it and to the evidence that measures it. It supersedes the scattered summaries of Tables 1.2, 3.2 and 7.2 for status purposes. Four verdicts are used. *Demonstrated* means the clause was measured and met under the documented conditions. *Partial* means supporting evidence exists but does not establish the clause as written. *Not established* means the required independent evidence is absent. *Out of scope* means the clause was explicitly deferred under the revised academic scope.

Table C.1: Requirement-to-evidence matrix.

| ID       | Clause                                                                            | Implementation and evidence                                                                                            | Verdict                              |
|----------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| REQ-01   | Commands remain within physical, rate and mask bounds at all times                | QP box constraints and rate rows; five nominal and 30 catalog cases pass the bounds, rate and friction checks          | Demonstrated in the catalog          |
| REQ-01   | Same clause during abrupt faults under fully saturated braking                    | Six exploratory cases log a healthy-channel rate override; section 8.1                                                 | Not met; see 8.1                     |
| REQ-02.1 | New actuator mask active within 10 ms                                             | Mask applied in the diagnosis sample on the 2 ms grid; 30 re-computed diagnostics                                      | Demonstrated as modeled              |
| REQ-02.2 | 90% of post-fault best-achievable output within 50 ms, then a 10% band for 100 ms | Affine-oracle diagnostics pass; normalized oracle gap $\leq 0.0898$ , section 6.9; no independent non-linear reference | Partial                              |
| REQ-02.3 | Healthy commands respect 40 kN/s and 400 deg/s                                    | Rate rows plus per-motor contribution checks in every run                                                              | Demonstrated in the catalog          |
| REQ-02.4 | Within 0.05 rad/s of nominal yaw rate for 200 ms after the fault                  | Maximum first-200-ms difference 0.013688 rad/s in the catalog; exceeded in the saturated exploratory cases             | Demonstrated in envelope only        |
| REQ-03a  | At least 60% of nominal straight-line deceleration after any one corner loss      | Minimum front-loss plateau 61.1147%; independent force-aligned static witness 61.2636%; sections 6.6 and 6.7           | Demonstrated at the evaluated states |
| REQ-03a  | Residual yaw moment $\leq 5\%$ of nominal yaw authority                           | Plateau hold moment 10.023 N m front, 2.663 N m rear against a 567.137 N m limit                                       | Demonstrated at the evaluated states |
| REQ-03b  | 60% of nominal road-wheel angle range and rate after one drive loss               | Functional common-rack bench; full range and 60% rate are modeling assumptions, not measured torque capability         | Functional model only                |
| REQ-03c  | DBBS yaw authority $\geq 15\%$ of nominal                                         | 16.6900% conservative frozen-condition witness under the 0.35 g cap; section 6.7                                       | Witness only, not universal          |

| ID         | Clause                                                                                                            | Implementation and evidence                                                                                     | Verdict                    |
|------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------|
| REQ-03c    | $\text{RMSE}(r) \leq 0.03 \text{ rad/s}$ , $ e_{lat}  \leq 0.50 \text{ m}$ , $a \leq 0.35 \text{ g}$ on the pulse | 8/8 cases pass; worst values 0.01146 rad/s, 0.2945 m, 0.2771 g; both centred directions retain the pass at 1 ms | Demonstrated               |
| REQ-04     | Architecture rationale, FMEA, FTA and dependent-failure analysis                                                  | F01–F18 register and D01–D08 dependencies, Appendix B; candidate HARA and FTTI budget, sections 7.2 and 7.3     | Qualitative rationale only |
| SG-04      | Prevent unintended or opposing rack torque                                                                        | No mitigation; F18 requires an independent monitor                                                              | Not addressed              |
| SG-05      | Detect and remove unintended corner brake force                                                                   | No mitigation; F17 requires independent force monitoring                                                        | Not addressed              |
| Extensions | Dual faults, standardized maneuvers, hardware in the loop                                                         | Structured as perspectives in Chapter 8                                                                         | Out of scope               |

Two entries deserve emphasis because they are easy to misread as successes. REQ-02.2 is supported only by an affine-oracle comparison, and section 7.3 shows that its 50 ms clause is unattainable by construction whenever the required brake-command change exceeds roughly 47% of a front corner’s range. REQ-03b passes a bench that measures the assumptions it was built from, so it verifies implementation consistency rather than steering capability under road load.

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